



AntiLeakBench: Preventing Data Contamination by Automatically Constructing Benchmarks with Updated Real-World Knowledge

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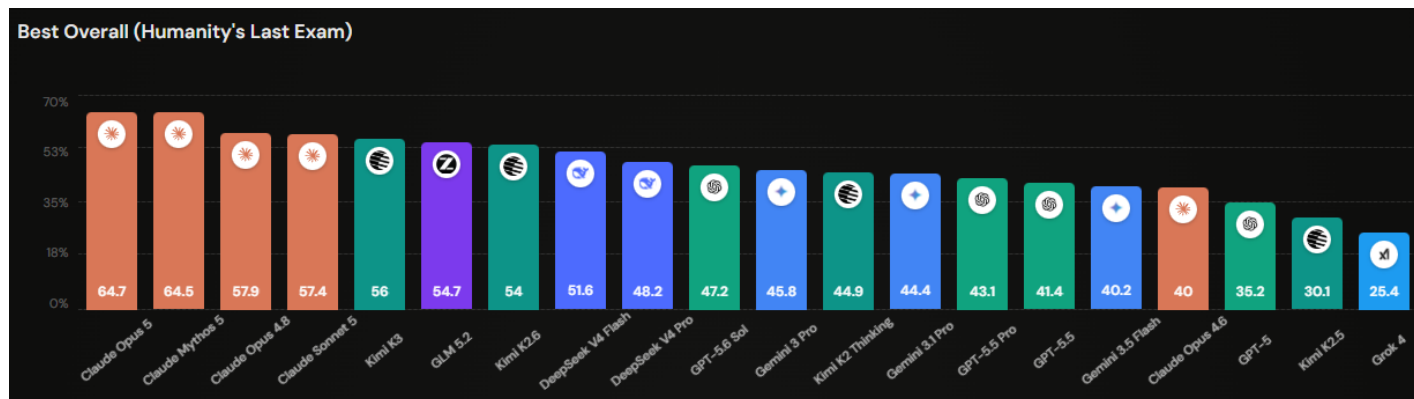
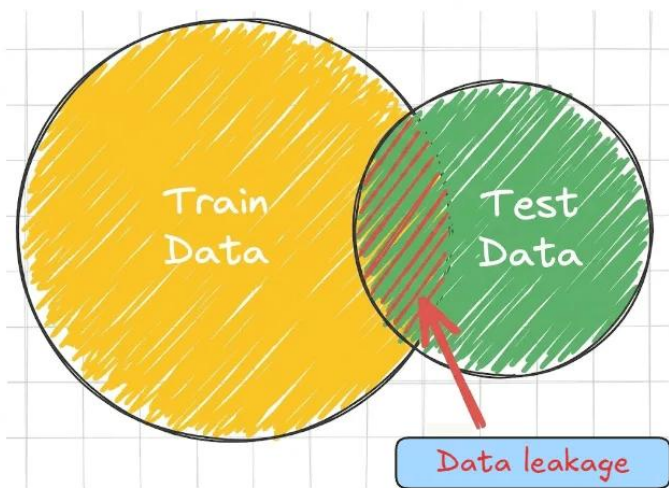
ACL 2025

SAC Highlights

HUMANE Lab
석사과정 김태균
2026.08.07

What is data contamination?

- Data contamination occurs when parts or characteristics of a test set leak into a training set
- If existing benchmarks are contaminated, then reliability of results is undermined



<https://blog.dailydoseofds.com/p/prevent-data-leakage-in-ml-pipelines>
<https://www.vellum.ai/llm-leaderboard>

Existing contamination-free benchmarks

- Collecting new data released after LLM's knowledge cutoff time
- But does not guarantee contamination risk
- It also requires human labor for maintenance



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How can we **automatically** construct
a **strictly** contamination-free benchmark?

AntiLeakBench: Data

- Wikidata -> (subject, relation, object)



Lionel Messi (Q615)

[Item](#) [Discussion](#)

Argentine association football player (born 1987)

[In more languages](#)

[Configure](#)

Language	Label	Description	Also known as
default for all languages	Lionel Messi	–	Lionel Andrés Messi Cuccitini Lionel Andrés Messi Cuccittini Lionel Andrés Messi Leo Messi Messi Lionel Andres Messi
English	Lionel Messi	Argentine association football player (born 1987)	
Korean	리오넬 메시	아르헨티나의 남자 축구 선수	메시

member of sports team

Paris Saint-Germain FC

start time August 2021
number of matches played/races/starts 58
acquisition transaction transfer
number of points/goals/set scored 22
end time 30 June 2023
sport number 30

[1 reference](#)

Inter Miami CF

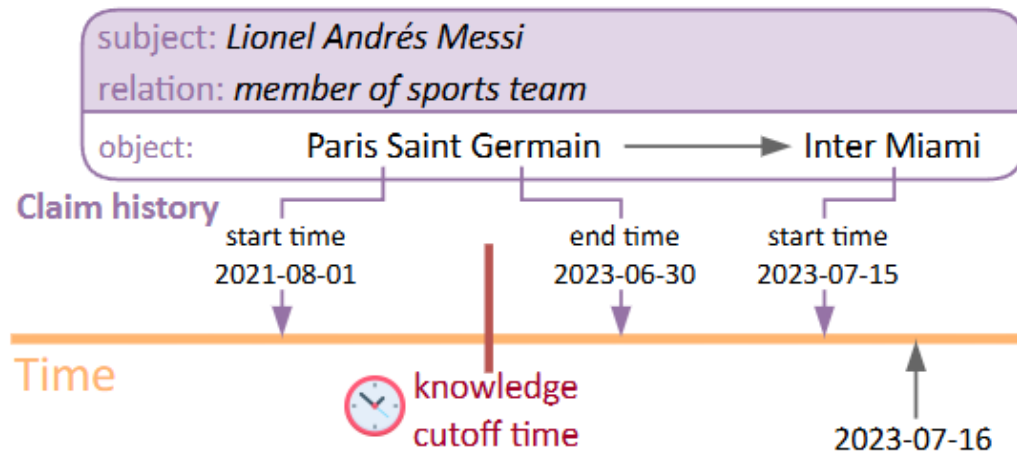
start time 15 July 2023
number of matches played/races/starts 67
number of points/goals/set scored 62
point in time 25 May 2026
sport number 10

(Lionel Messi, member of sports team, Paris Saint Germain)

AntiLeakBench: Updated knowledge

- Identify a fact whose object changes after the model's knowledge cutoff

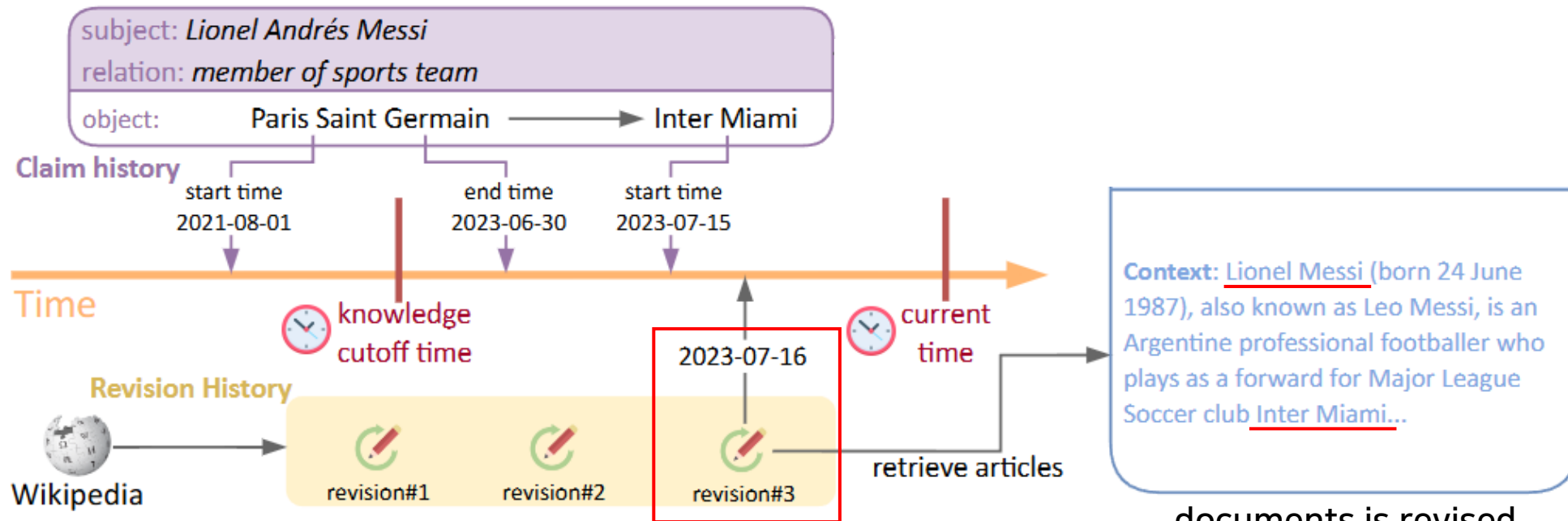
① Identify Updated Knowledge



AntiLeakBench: Supporting Documents

- Find the Wikipedia page revision made after the start time of the updated knowledge

① Identify Updated Knowledge



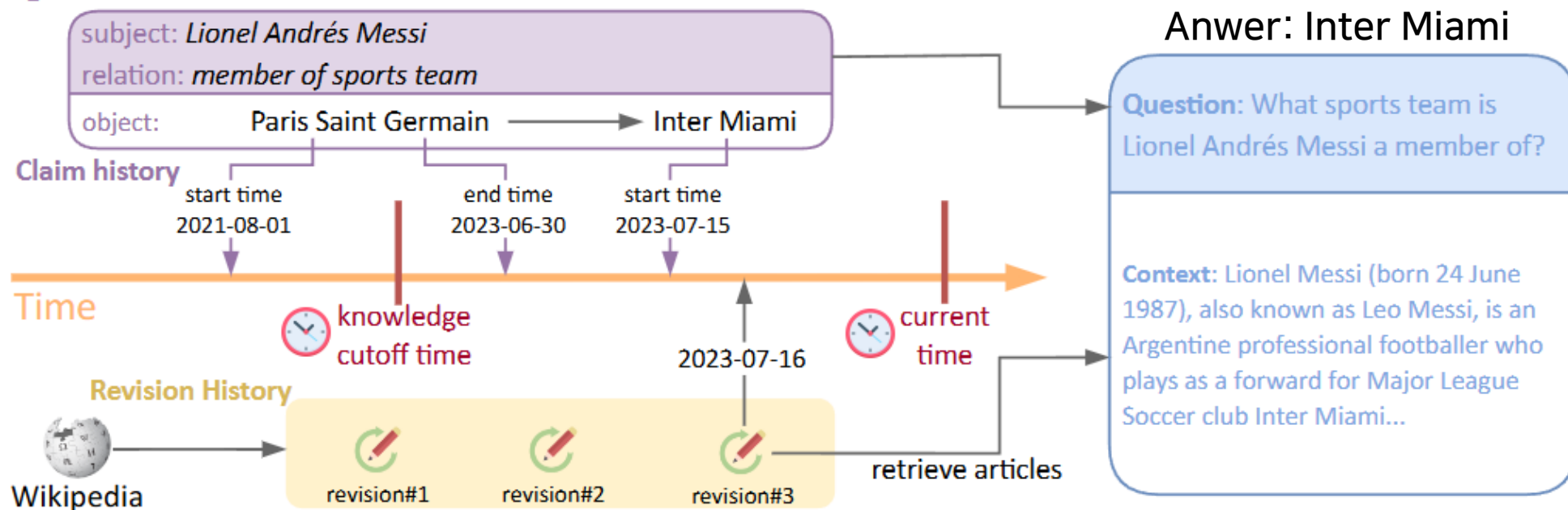
② Build Supporting Documents

documents is revised after LLM's cutoff time

AntiLeakBench: Constructing samples

- Construct QA samples: (Q, C, A)

① Identify Updated Knowledge



② Build Supporting Documents

③ Construct Contamination-Free Samples

AntiLeakBench: Task

- QA
 - Single-hop: directly asks for the updated fact
 - Multi-hop: requires chaining multiple claims

Question: Who is the head coach of the sports team that Lionel Andrés Messi is a member of?

Context:

Passage 1: Lionel Messi (born 24 June 1987), also known as Leo Messi, is an Argentine professional footballer who plays as a forward for Major League Soccer club Inter Miami...

Passage 2: Gerardo Daniel Martino (born 20 November 1962) is an Argentine professional football coach and former player, who is currently the head coach of Major League Soccer club Inter Miami...

Answer: Gerardo Daniel Martino

AntiLeakBench: Question formats

- Generation
- Multiple-choice
 - Option: correct/outdated/noise/unknown

Attributes	Examples
question (generation)	What sports team is Lionel Andrés Messi a member of?
answer (generation)	Inter Miami CF Inter Miami Club Internacional de Fútbol Miami
question (multi-choice)	What sports team is Lionel Andrés Messi a member of? A. Inter Miami CF B. Paris Saint-Germain F.C. C. Prime Minister of Romania D. Unknown.
answer (multi-choice)	A

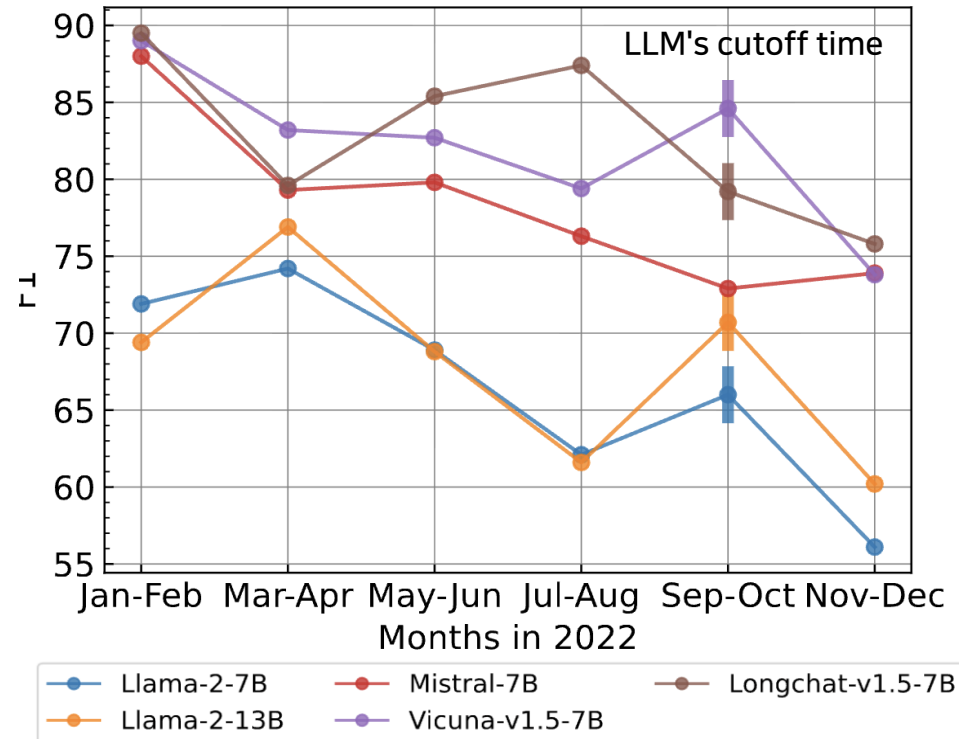
Experiments

- Investigate the impact of data contamination on model performance
 - Compare 12 LLMs grouped by knowledge cutoff date
 - Divide samples into pre-cutoff and post-cutoff groups
- Metric: EM, F1, ACC

Model	Release time	Knowledge cutoff time
Llama-2-7B	2023-07	2022-09
Llama-2-13B	2023-07	2022-09
Mistral-7B	2023-09	2022*
Vicuna-v1.5-7B	2023-07	2022-09
Longchat-v1.5-7B	2023-07	2022-09
Llama-3.1-8B	2024-07	2023-12
Phi-3.5-mini	2024-08	2023-10
Qwen-2-7B	2024-06	2023*
Mistral-Nemo-12B	2024-07	2024-04
Gemma-2-9B	2024-08	2024-06*
GPT-4o-mini	2024-07	2023-10
GPT-4o	2024-07	2023-12

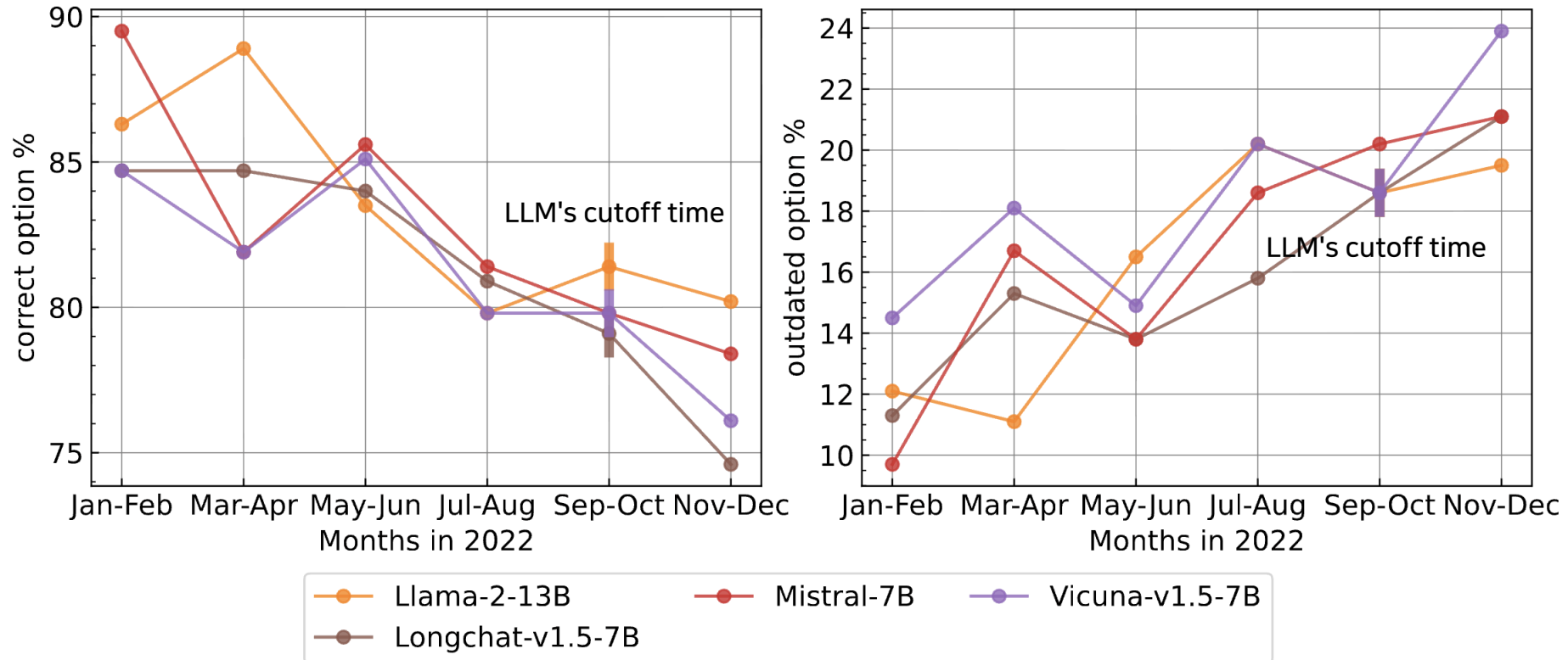
Data contamination analysis

- Performance decline for most LLMs after their knowledge cutoff time
 - Pre-cutoff samples inflate LLMs' performance



Data contamination analysis

- LLMs increasingly favor outdated options over correct ones



Overall performance analysis

- Proprietary models lead in performance
- Multi-hop tasks are more challenging

Language Models	Single-Hop								Multi-Hop								Avg	
	Gold		$N_d=3$		$N_d=5$		$N_d=7$		Gold		$N_d=3$		$N_d=5$		$N_d=7$			
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-2-7B	40.6	63.5	16.8	41.2	11.6	30.9	9.4	24.5	33.6	50.2	19.4	32.2	15.8	28.1	12.2	22.7	19.9	36.7
Llama-2-13B	42.7	65.3	14.0	40.6	9.4	30.6	7.0	24.0	13.3	34.6	4.1	21.5	2.7	17.8	2.3	15.2	11.9	31.2
Mistral-7B	65.4	77.2	27.8	41.3	16.7	27.3	7.3	15.3	21.4	27.9	11.5	17.2	8.1	14.3	6.5	11.1	20.6	29.0
Vicuna-v1.5-7B	66.8	79.9	39.1	60.4	25.8	48.3	15.3	39.1	26.0	43.5	11.1	22.9	8.1	19.5	5.4	15.7	24.7	41.2
Longchat-v1.5-7B	75.5	84.5	58.2	72.8	47.6	65.5	37.0	56.3	38.8	51.4	17.6	30.6	12.0	25.8	4.7	3.9	36.4	48.9
Llama-3.1-8B	19.2	66.2	21.4	59.4	18.1	53.5	14.2	45.7	24.4	50.2	11.7	33.0	9.4	27.5	6.8	21.9	15.6	44.7
Phi-3.5-mini	69.0	78.7	34.0	40.5	26.5	33.7	15.2	22.2	45.4	59.7	20.8	29.5	14.9	21.1	9.8	14.4	29.4	37.5
Qwen-2-7B	54.8	72.4	15.5	38.5	9.8	26.6	7.2	21.2	35.9	48.3	23.7	33.4	18.1	26.1	13.6	20.1	22.3	35.8
Mistral-Nemo-12B	82.7	89.7	75.6	83.8	66.3	75.1	51.8	62.2	57.7	67.3	39.1	47.7	33.8	41.4	24.0	29.0	53.9	62.0
Gemma-2-9B	85.0	91.6	80.2	86.2	68.8	75.2	55.4	61.2	82.7	86.4	63.0	68.3	55.8	61.2	49.0	53.5	67.5	73.0
GPT-4o-mini	78.5	88.1	80.3	89.2	79.1	88.1	79.2	88.5	68.8	83.1	60.5	75.3	57.1	73.1	54.2	70.6	69.7	82.0
GPT-4o	81.2	89.5	84.1	90.8	83.5	90.3	84.8	91.4	71.5	85.9	71.9	86.1	70.2	84.8	70.2	84.8	77.2	87.9

Table 4: EM (Exact Match) and F1 results in the **generation** format on AntiLeakBench. Gold means only gold documents; N_d is the number of distracting documents. The best is in **bold**.

Takeaways

- Performance declines after the knowledge cutoff, while outdated answers become more likely

Opinion

- award 이유
- 한계 및 개선 지점